

How much hardware to run campus AI chat

Sizing GPU capacity for a University-wide conversational AI service, from realistic daily peaks up to a quarter of the entire campus using it at the same moment, and what the incoming HGX B300 system adds.

Prepared 2026-08-01 · MindRouter AI Gateway · Model under test: gemma-4-31b · H200 figures from live load simulation; B300 figures projected from vendor specifications

One H200 GPU serves about **170 concurrent chat users** within the target Service Level Objectives. Grounded in enterprise adoption patterns, realistic peak concurrency for the 18,000-person campus runs from roughly **180** at a quiet launch to about **1,800** during an exceptional exam-week surge, needing on the order of **1 to 11 H200s**. The incoming **HGX B300** node covers that entire range with headroom, delivering about the capacity of **24 H200s** in one chassis. A quarter-campus simultaneous event (**4,500**), well beyond realistic demand, would need about 26 H200s or the B300 node at its edge.

PER H200 · SUSTAINED ~170 concurrent users within SLO	HEAVY ADOPTION PEAK ~5-8 H200s · 720 to 1,260 concurrent	EXCEPTIONAL EVENT ~8-11 H200s · 1,400 to 1,800 concurrent	ONE HGX B300 NODE ~4,100 covers every realistic peak
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1 Users, and the Service Level Objectives

Capacity is quoted in **concurrent users**, meaning people *actively in a conversation* (reading, typing, or waiting on a reply) in the same brief moment, not accounts or daily visitors. This distinction is the whole story, because a person spends most of a chat session reading and thinking, with the model actively generating tokens for only a small slice of that time. One GPU therefore serves many *people* because they are rarely all mid-response together.

"Comfortable service" is not a vague adjective. We hold the service to five concrete **Service Level Objectives (SLOs)**, below; capacity is the largest number of concurrent users at which **all five** still hold. Loosening the SLOs raises the user count on the same hardware; tightening them lowers it, so every figure in this briefing is stated against this specific bar.

SERVICE LEVEL OBJECTIVE	THRESHOLD	WHAT THE USER EXPERIENCES
Time to first token (TTFT)	p95 ≤ 2.0 s	The reply starts almost immediately, even on a worst-case (95th-percentile) request.
Streaming throughput	p10 ≥ 15 tokens/s	Even the slowest 10% of responses stream faster than a person reads (~1 token ≈ ¼ word).
Error rate	≤ 1%	Requests essentially never fail outright.
Inter-token stalls	≤ 2% of responses	No visible mid-response freeze (no >2 s gap between tokens).
End-to-end response time	p95 ≤ 140 s	Even long answers finish in a reasonable window.

These thresholds are interactive-chat norms: a first token within ~2 seconds reads as responsive, and 15 tokens/second comfortably outpaces reading. Percentiles (p95, p10) hold the SLO at the tail, not just the average, so a bad-luck request is still acceptable. The SLOs are policy choices the University can adjust, and the capacity figures scale with them.

2 Likely adoption and concurrent usage

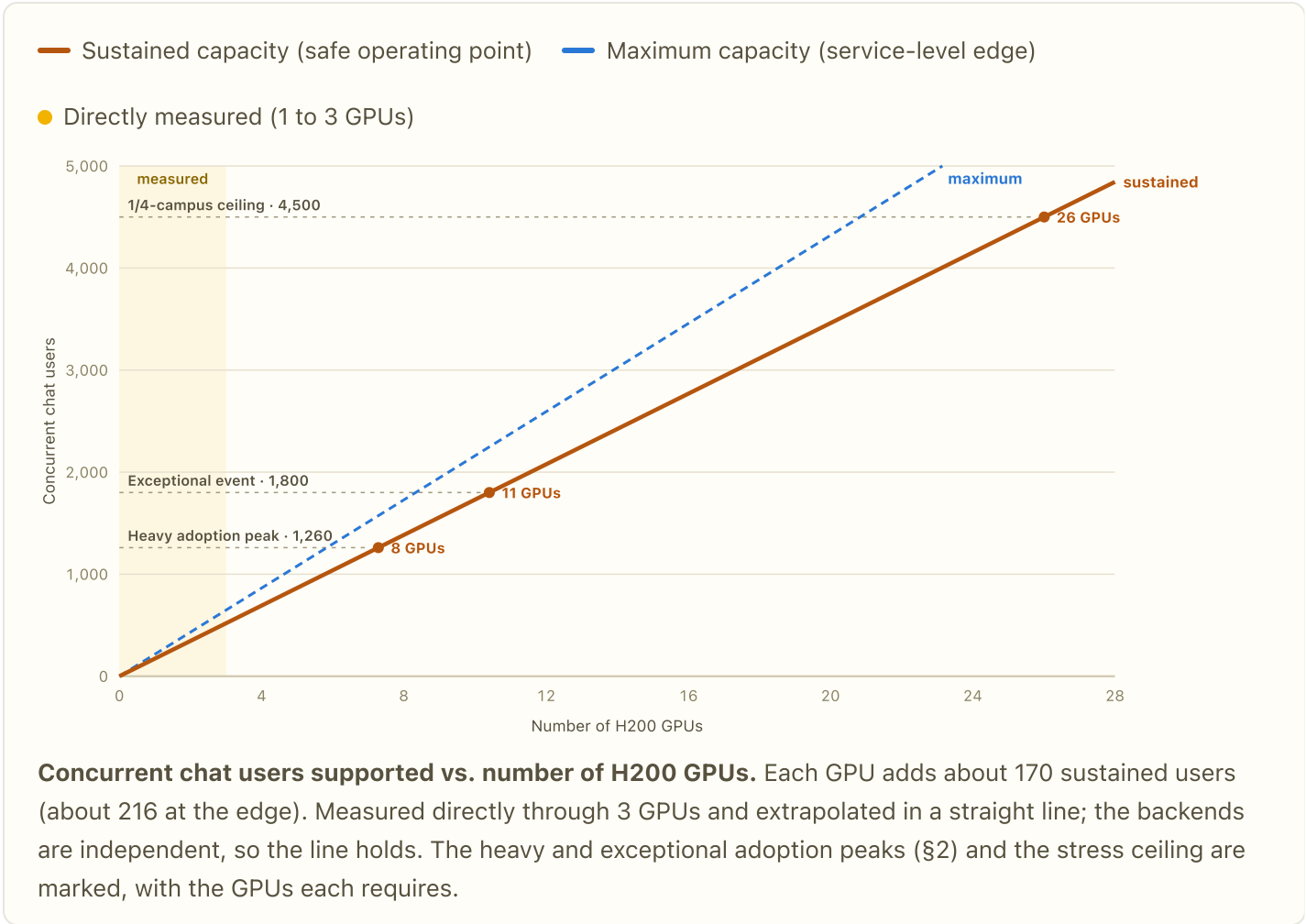
The hardware requirement hinges on how many people are chatting in the same instant, which is the product of two things: **adoption** (what share of campus uses the tool at all) and **concurrency** (what fraction of them are mid-request in the same moment). Active users far outnumber concurrent users, because a person spends most of a session reading and typing, not waiting on tokens. Drawing on published adoption and concurrency patterns for enterprise SaaS, learning-management systems, and AI chat rollouts, the table below is the likely demand curve for an 18,000-person campus. Peak concurrency is expressed as a share of the full population.

ADOPTION SCENARIO	ACTIVE USERS	PEAK CONCURRENT	H200S · SUSTAINED	SHARE OF ONE HGX B300 NODE
Light usage	20% (3,600)	1 to 2% (180 to 360)	1 to 2	~4 to 9%
Moderate usage	40% (7,200)	2 to 4% (360 to 720)	2 to 4	~9 to 18%
Heavy campus adoption	70% (12,600)	4 to 7% (720 to 1,260)	5 to 8	~18 to 31%
Exceptional event (exam week, campus announcement)	80% or more	8 to 10% (1,400 to 1,800)	8 to 11	~34 to 44%

The takeaway for planning: even an exceptional event, the busiest realistic moment (exam week or a campus-wide announcement), peaks near **1,800 concurrent users**. That is served by roughly 8 to 11 H200s, or by **less than half of a single HGX B300 node**. The entire realistic adoption curve, from a quiet launch through heavy sustained use to an exceptional spike, fits comfortably inside the incoming B300 system with capacity to spare.

The quarter-campus figure in context. A quarter of the whole population (4,500 people) chatting in the same instant is roughly **2.5 times the most extreme realistic peak**. We carry it only as a stress ceiling, the point at which the service would still never say no, not as expected demand. It would require about 26 H200s, or a single HGX B300 node run to its maximum.

3 Capacity scales in a straight line with GPUs



Because each GPU runs an independent copy of the model, total capacity is simply the per-GPU number times the GPU count, a straight line. We confirmed this by measuring one, then three GPUs live: three delivered essentially triple the capacity of one, and the AI gateway that routes traffic across them kept pace without becoming the bottleneck (see §5). Extrapolating that line gives the GPU counts above.

4 The HGX B300 upgrade: far more per box

The incoming HGX B300 system (eight B300 GPUs joined by NVIDIA's fastest interconnect) is a generational step up from the H200. Two effects compound. First, the hardware itself is much stronger: about 3.5 times the raw math throughput, twice the on-board memory (288 GB per GPU vs 141 GB), and much higher memory bandwidth. Second, because the B300 uses NVIDIA's Blackwell architecture, we can run this model in **NVFP4**, a new four-bit number format that roughly halves the model's memory footprint and doubles its math throughput compared with the FP8 format the H200 uses, while a per-block scaling technique preserves answer quality.

For chat, the metric that matters is how fast the GPU generates words for many users at once, and NVIDIA's published measurements put the HGX B300 at about **3.4 times** the H200 on exactly that. Applied conservatively to our own measured H200 numbers, this puts a single B300 GPU at roughly **three times** the chat capacity of an H200, with the doubled memory giving it ample room to hold many more simultaneous conversations.

	H200 (FP8)	B300 (NVFP4)	ADVANTAGE
On-board memory	141 GB	288 GB	2×
Number format	FP8 (8-bit)	NVFP4 (4-bit)	½ the footprint
Sustained users, per GPU	~170	~510	~3×
Sustained users, full 8-GPU node	~1,360	~4,100	~3×
Maximum users, full 8-GPU node	~1,730	~5,200	~3×

What the single HGX B300 delivers. One node sustains roughly **4,100 concurrent chat users** and peaks near **5,200**, about the capacity of **24 H200 GPUs** in one chassis. That covers the typical busy-hour peak more than four times over, absorbs a heavy exam-season surge with wide headroom, and on its own reaches the extreme quarter-campus ceiling that would otherwise require roughly 26 separate H200s. In practical terms, the purchase provisions the entire realistic demand range with room to spare, from a single well-connected system that is simpler to operate than a large fleet of discrete GPUs.

One caveat of judgment: the B300 figures are engineering projections from vendor specifications and published benchmarks, not our own measurements the way the H200 numbers are. On delivery we will run the same live simulation against the B300, and validate that NVFP4 holds answer quality for student-facing use (a standard calibration and evaluation step), before committing the numbers to planning.

5 How we know, and what we improved

Measured, not estimated

We built a simulator that behaves like real people: multi-turn conversations, realistic pauses to read and type, and a genuine spread of short questions and long pasted documents. We then raised the number of simulated users against live GPUs until a quality target broke, the point where capacity ends. Every H200 figure here comes from that live testing on production hardware, cross-checked against the GPUs' own internal metrics.

We removed the software ceiling

Early testing found the routing gateway, not the GPUs, capping capacity and failing hard under load. A round of engineering (faster request handling, smarter streaming, shared capacity accounting) **more than tripled** what the gateway sustains and changed its overload behavior from abrupt failure to graceful slowdown. The gateway is no longer the limit; the GPUs are, which is exactly what lets capacity scale in a straight line.

Bottom line for planning. The incoming HGX B300 node alone covers the entire realistic adoption curve, through an exceptional exam-week event, with capacity to spare, and reaches the quarter-campus stress ceiling at its edge. If provisioning with H200s instead, budget for roughly **5 to 11 GPUs** to cover heavy adoption through an exceptional peak, hold the **~26-GPU** figure as the extreme ceiling, and add one or two GPUs for redundancy so a single failure never drops service. Either way, capacity grows incrementally: add hardware as adoption rises, and the system scales with it.

6 Assumptions and caveats

- **Population:** 18,000 students and employees; scenario percentages are of that total. A quarter-campus ceiling is 4,500 concurrent users.
- **Model and hardware:** gemma-4-31b on NVIDIA H200 (FP8) and, projected, NVIDIA B300 (NVFP4). A different model or GPU changes the per-GPU number and would need its own measurement.
- **Workload:** general chat (questions, drafting, explanations). Heavy document analysis or retrieval-augmented use is more demanding per request and would lower the per-GPU user count.
- **H200 extrapolation:** per-GPU capacity is measured through 3 GPUs and extended in a straight line. The linearity is well-founded (independent GPUs, a gateway proven to keep pace), but the largest figures are projections, not measurements.
- **B300 projection:** based on vendor specifications and published token-generation benchmarks (about 3.4× the H200), applied conservatively at 3×. To be validated on delivery, along with NVFP4 answer quality.
- **Adoption and concurrency:** the adoption tiers and peak-concurrency ranges (§2) are drawn from published enterprise SaaS, LMS, and AI-chat deployment patterns, expressed as a share of the 18,000-person population. They are planning estimates; actual provisioning should track observed usage once the service is live, starting near the low end and growing with demand.
- **Service level:** all figures assume the five quality targets in §1; relaxing them raises the per-GPU user count.